

Formal Methods and Specification (SS 2025/26)

Lecture 2: Proofs

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Checking Correctness of Programs

Example: You write a program for a friend

Specification:

- ▶ Input: $x \in \mathbb{Z}$
- ▶ Output: $\hat{x} \in \mathbb{Z}, \hat{x} < x - |f(x)|$

Program:

```
if  $f(x) \geq 0$  then                                //  $f$  behaves like a mathematical function
    return  $x - f(x) - 1$ 
else
    return  $x + f(x) - 1$ 
```

Friend: Is it correct? For input -3 , yes! $8? -2? \dots$ testing

General situation in life

Alternatively: evidence in form of argument

Problems: Which arguments allowed? programming language specific

Reduction to Logic

Specification:

- ▶ Input: $x \in \mathbb{Z}$
- ▶ Output: $\hat{x} \in \mathbb{Z}$, $\hat{x} < x - |f(x)|$

Program:

```
if  $f(x) \geq 0$  then  
    return  $x - f(x) - 1$   
else  
    return  $x + f(x) - 1$ 
```

Program is correct iff *the verification condition*

$$\forall x \in \mathbb{Z}. \quad \begin{array}{c} f(x) \geq 0 \Rightarrow x - f(x) - 1 < x - |f(x)| \\ \wedge \\ \neg f(x) \geq 0 \Rightarrow x + f(x) - 1 < x - |f(x)| \end{array}$$

holds (independent of f , using properties of $\mathbb{Z}$).

Similar problem: $\mathbb{Z}$ is an infinite set

How to convince somebody with absolute **certainty**?

Proofs

Proof: **Absolutely certain evidence** that a formula holds, that **everybody** can check **always** and **everywhere**.

Proofs that are written formally,
can even be checked for correctness by a computer
(viz. proof-carrying code)

“formula ϕ holds”: ϕ is a **logical consequence** of Φ ($\Phi \models \phi$)
where Φ are the things we already know.

Often we will not state Φ explicitly, and write $\models \phi$.

Recall: validity of a formula

Proof by Equivalence

One can prove $A \Leftrightarrow B$ by

proving that A is equivalent to B based on equivalence rules
(see MI-TES, Section 1.5 in lecture notes)

Example: $A \vee [B \wedge C]$ is equivalent to $[A \vee B] \wedge [A \vee C]$

Then one can prove ϕ by proving $\phi \Leftrightarrow \top$

But unfortunately, this is **not enough** (the method is not complete).

Today: a **complete** proof method:

There is a proof for every formula that holds.

outermost symbol: root of syntax tree

$$[p \Rightarrow q] \Rightarrow [[p \wedge r] \Rightarrow [q \wedge r]]$$

Example

*You will find the animated version of the example in the video.
In written form, you can find it on the next slide.*

Example

Example: Prove

$$[p \Rightarrow q] \Rightarrow [[p \wedge r] \Rightarrow [q \wedge r]]$$

We assume $p \Rightarrow q$ and prove $[p \wedge r] \Rightarrow [q \wedge r]$. For proving $[p \wedge r] \Rightarrow [q \wedge r]$ we assume $p \wedge r$ and prove $q \wedge r$. For proving $q \wedge r$ we prove both q and r :

- ▶ Proof of q : From the assumption $p \wedge r$ we conclude both p and r . From the assumptions p and $p \Rightarrow q$ follows q , which finishes this part of the proof.
- ▶ Proof of r : We already know that from the assumption $p \wedge r$ follows r , which finishes also this part of the proof.

Proof Method

For proving something, we keep a **list** of things (assumptions) that we already **know**, and

- ▶ **add known** things (“to assume, to conclude”), and
- ▶ **simplify** the thing to **prove**,

until what we want to prove is known.

As usual we assume an infinite set of predicate and function symbols

We handle types intuitively.

Proof Rules—Propositional Logic without Negation

A , B meta-variables, i.e., variables that represent arbitrary formulas

The rules refer to the **outer-most symbol** of a formula

How to work on things that we have to prove?

$A \wedge B$: Separately prove A and B

$A \vee B$: Assume $\neg A$ and prove B (or vice versa).

$A \Rightarrow B$: Assume A and prove B .

How to generate new knowledge?

$A \wedge B$: Conclude both A and B

$A \vee B$: Do a case distinction: first write "Case A :", assume A and finish the proof. Then write "Case B :", assume B , and finish the proof.

$A \Rightarrow B$: If we also know A then we conclude B .

What about $\Leftrightarrow$?

$A \Leftrightarrow B$: shortcut for $[A \Rightarrow B] \wedge [B \Rightarrow A]$

Proof Assistant

`https://fme-proof-assistant.netlify.app/`
prototype!

Proof by Contradiction

Example:

$$[p \wedge \neg p] \Rightarrow q$$

Holds, due to left-hand side $p \wedge \neg p$

To prove the formule, we assume $p \wedge \neg p$ and prove q . From the assumption $p \wedge \neg p$ we know both p and $\neg p$, which is a

Contradiction: We know both a formula and its negation.

A contradiction finishes **every** proof (**successfully!**)

So: this proves what we originally wanted to prove!

Further example: $p \Rightarrow [\neg p \Rightarrow q]$

Attention: The rules for
proving conjunction and using disjunction in an assumption
start two **separate** proofs.

Negations: Example

*You will find the animated version of the example in the video.
In written form, you can find it on the next slide.*

Example

$$[[p \Rightarrow q] \wedge [p \Rightarrow \neg q]] \Rightarrow \neg p$$

We assume $[p \Rightarrow q] \wedge [p \Rightarrow \neg q]$ and prove $\neg p$. For proving $\neg p$ we assume p and try to find a contradiction.

From the assumption $[p \Rightarrow q] \wedge [p \Rightarrow \neg q]$ we know the assumptions $p \Rightarrow q$ and $p \Rightarrow \neg q$. From the assumptions p and $p \Rightarrow q$ follows q , and from the assumptions p and $p \Rightarrow \neg q$ follows $\neg q$. So we have both assumptions q and $\neg q$, which is a contradiction. This finishes the proof.

Proof Rules—Propositional Case

How to work on things that we have to prove?

$A \wedge B$: Separately prove A and B

$A \vee B$: Assume $\neg A$ and prove B (or vice versa).

$A \Rightarrow B$: Assume A and prove B .

$\neg A$: Assume A , and try to find a contradiction

How to generate new knowledge?

$A \wedge B$: Conclude both A and B

$A \vee B$: Do a case distinction: first write "Case A :", assume A and finish the proof. Then write "Case B :", assume B , and finish the proof.

$A \Rightarrow B$: If we also know A then we conclude B .

Quantifiers: Reminder

For us, the scope of quantifiers is as far to the right as possible!

So

$$\exists x . P(x) \wedge Q(x)$$

means

$$\exists x . [P(x) \wedge Q(x)],$$

and not

$$[\exists x . P(x)] \wedge Q(x).$$

Universal Quantifiers: Example

Assumptions:

Assumptions:

$$f(a) \geq 0$$

To prove:

$$a - f(a) - 1 < a - |f(a)|$$

Assumptions:

$$\neg f(a) \geq 0$$

To prove:

$$a + f(a) - 1 < a - |f(a)|$$

To prove:

$$\begin{array}{l} \forall x \in \mathbb{Z}. \quad \begin{array}{l} f(x) \geq 0 \Rightarrow x - f(x) - 1 < x - |f(x)| \\ \quad \quad \quad \wedge \\ \neg f(x) \geq 0 \Rightarrow x + f(x) - 1 < x - |f(x)| \end{array} \\ f(a) \geq 0 \Rightarrow a - f(a) - 1 < a - |f(a)| \\ \quad \quad \quad \wedge \\ \neg f(a) \geq 0 \Rightarrow a + f(a) - 1 < a - |f(a)| \end{array},$$

where a is new (“arbitrary, but fixed”).

Universal Quantifiers: Example

$$\begin{array}{c} f(x) \geq 0 \Rightarrow x - f(x) - 1 < x - |f(x)| \\ \forall x \in \mathbb{Z}. \quad \quad \quad \wedge \\ \neg f(x) \geq 0 \Rightarrow x + f(x) - 1 < x - |f(x)| \end{array}$$

We want to prove the formula for an arbitrary integer, that is, an integer, about which we do **not know anything**. So we introduce a **new** constant a , and prove

$$\begin{array}{c} f(a) \geq 0 \Rightarrow a - f(a) - 1 < a - |f(a)| \\ \quad \quad \quad \wedge \\ \neg f(a) \geq 0 \Rightarrow a + f(a) - 1 < a - |f(a)| \end{array}$$

To prove the conjunction we prove both

$$f(a) \geq 0 \Rightarrow a - f(a) - 1 < a - |f(a)|$$

and

$$\neg f(a) \geq 0 \Rightarrow a + f(a) - 1 < a - |f(a)|.$$

Universal Quantifiers: Example

To prove

$$f(a) \geq 0 \Rightarrow a - f(a) - 1 < a - |f(a)|,$$

we assume

$$f(a) \geq 0$$

and prove

$$a - f(a) - 1 < a - |f(a)|$$

which clearly holds under this assumption.

The proof of

$$\neg f(a) \geq 0 \Rightarrow a + f(a) - 1 < a - |f(a)|$$

is similar.

Existential Quantifiers: Example

$$\exists x \in \mathbb{N} . x \geq 4 \wedge 2 \nmid x$$

Proving that something exists that fulfills a certain property:
find one object that fulfills this property.

For example, here: 7

Substitute, the result is $7 \geq 4 \wedge 2 \nmid 7$

Clearly both $7 \geq 4$ and $2 \nmid 7$.

Proof Rules Including Quantifiers 1

How to work on things that we have to prove?

$\exists x . A$: Choose (by intuition, creativity, or any other special connection with god) a term t , and prove $A[x \leftarrow t]$

$\forall x . A$: Choose a **new** constant, e.g., a , write/say “let a be arbitrary but fixed” and prove $A[x \leftarrow a]$, instead

$A \wedge B$: Separately prove A and B

$A \vee B$: Assume $\neg A$ and prove B (or vice versa).

$A \Rightarrow B$: Assume A and prove B .

$\neg A$: Assume A , and try to find a contradiction

Remember the definition of the notion “term”.

Especially: Every constant is a term

(as usual, we assume that there are infinitely many constants).

Proof Rules Including Quantifiers 2

How to generate new knowledge?

- $\exists x . A$: Choose a **new** constant, e.g., a , write/say "let a be such that $A[x \leftarrow a]$ ", and add $A[x \leftarrow a]$ to our knowledge base
- $\forall x . A$: Choose (by intuition, creativity, or any other special connection with god) a term t and conclude $A[x \leftarrow t]$
- $A \wedge B$: Conclude both A and B
- $A \vee B$: Do a case distinction: first write "Case A :", assume A and finish the proof. Then write "Case B :", assume B , and finish the proof.
- $A \Rightarrow B$: If we also know A then we conclude B .

Example

*You will find the animated version of the example in the video.
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Example

$$[[\forall x. P(x) \Rightarrow Q(x)] \wedge [\exists x. P(f(x))]] \Rightarrow [\exists x. Q(x)]$$

(scope of quantifiers as far as possible!)

We assume $[\forall x. P(x) \Rightarrow Q(x)] \wedge [\exists x. P(f(x))]$, that is, we assume both $\forall x. P(x) \Rightarrow Q(x)$ and $\exists x. P(f(x))$. We want to prove $\exists x. Q(x)$.

Due to the assumption $\exists x. P(f(x))$ we can choose a new constant a such that $P(f(a))$. Due to the assumption $\forall x. P(x) \Rightarrow Q(x)$ we can conclude $P(f(a)) \Rightarrow Q(f(a))$. For proving $\exists x. Q(x)$ we choose for x the term $f(a)$, and now it suffices to prove $Q(f(a))$, which follows from the assumptions $P(f(a))$ and $P(f(a)) \Rightarrow Q(f(a))$. This finishes the proof.

Proof Rules for Equalities

For every term t we can assume $t = t$.

If we know $t_1 = t_2$,
then we can always substitute t_1 for t_2 and vice versa.

Example:

If we know $a = b$ and $P(a)$,
then we can add the assumption $P(b)$.

But: see slide “Substitutions: Avoiding Name Clashes” below.

Using Equivalences

It is always allowed to **replace equivalences**
(e.g., $A \wedge B$ is equivalent with $B \wedge A$)

Strictly necessary is only the equivalence between **A** and **$\neg\neg A$**

Especially: When proving A , we can

1. replace it by $\neg\neg A$,
2. apply the rule for proving negations,
3. and add $\neg A$ as additional knowledge.

Lemmas: Example:

*You will find the animated version of the example in the video.
In written form, you can find it on the next slide.*

Lemmas: Example:

$$\left[p \wedge \left[\left[p \vee q \right] \Rightarrow r \right] \right] \Rightarrow r$$

We assume p , $\left[p \vee q \right] \Rightarrow r$ and try to prove r .

Now it would be useful to know $p \vee q$,
which would be easy to prove from p .

This is possible, but needs a separate proof (lemma).

Proof of the lemma $p \vee q$ from our current knowledge: For proving $p \vee q$ we assume $\neg q$, and prove p . The formula p is already a known fact, so the proof of the lemma is finished.

Now we add the formula $p \vee q$ to our know facts. From the formula $\left[p \vee q \right] \Rightarrow r$ we know r which finishes the main proof.

Lemmas in General

We also have only **one** formula to prove,
but we may have **several assumptions**.

If we would like to **prove** some **additional knowledge**,
we can prove it in a separate proof (from the current knowledge)
we can add to our knowledge.

A lemma may recursively contain further lemmas etc

Usage of Lemmas: Typical Examples

If we know $A \Rightarrow B$, then
prove A using a lemma, and
then from A and $A \Rightarrow B$ we can conclude B .

$A \vee \neg A$ (case distinction) can be proved without any assumptions,
so we can add formulas of this form **always**.

If we know $\neg A$,
prove A using a lemma,
contradiction, successful proof.

In all cases:
 A and B can be complicated formulas!

Attention: A lemma is a **separate** proof!
Except for the formula that we prove in the lemma,
no influence on the main proof.

Example:

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Example:

$$[\neg \forall x . P(f(g(x)))] \Rightarrow [\exists x . \neg P(x)]$$

We assume $\neg \forall x . P(f(g(x)))$ and prove $\exists x . \neg P(x)$. For proving an existential quantifier, we would need a term to substitute, but we do not know anything useful. So we add a double negation, resulting in $\neg \neg \exists x . \neg P(x)$, and apply the rule for proving negations which adds $\neg \exists x . \neg P(x)$ as additional knowledge.

Now we have two formulas with negations as knowledge. If we would know $\forall x . P(f(g(x)))$, we would arrive at a contradiction. Hence we prove this formula in a lemma.

Lemma: Proof of $\forall x . P(f(g(x)))$

In the lemma, we apply the proof rule for universal quantifiers, and prove $P(f(g(a)))$ for a new, arbitrary, but fixed constant a . Now we do not have any proof rule left to apply. So we again add a double negation, and then add $\neg P(f(g(a)))$ as additional knowledge.

At this point, we are still proving the lemma. We know $\neg \exists x . \neg P(x)$ and $\neg P(f(g(a)))$. The latter formula contains the term $f(g(a))$ that we may use to prove an existential quantifier. Hence, we prove $\exists x . \neg P(x)$ using a another lemma within the current lemma.

Example continued

The proof of this second lemma immediately succeeds by choosing $f(g(a))$ for x . Hence we can add $\exists x . \neg P(x)$ to the knowledge in the original lemma. This creates a contradiction which proves this original lemma. As a consequence, we can add $\forall x . P(f(g(x)))$ to the main proof. This again creates a contradiction which finishes the proof.

Substitution: Avoiding Name Clashes

Attention: Substitutions must **not** result in **new bound variables**!

Example: If we want to prove

$$\exists x \forall y . p(x, z)$$

we cannot choose $f(y)$ for the variable x , since

$$[\forall y . p(x, y)][x \leftarrow f(y)] \text{ is } \forall y . p(f(y), z),$$

where y is suddenly bound by a quantifier .

Example: If we know both $\exists x . P(x, y)$ and $y = f(x)$, then

we are not allowed to substitute $f(x)$ for y in $\exists x . P(x, y)$

since the result $\exists x . P(x, f(x))$ would have a new bound variable.

Solution: Rename the bound variable: $\exists z . P(z, y)$ (always allowed)

Unsuccessful Proofs: Examples

$$p \vee [\neg p \wedge q]$$

Counter-example: $\{p \mapsto \perp, q \mapsto \perp\}$

$$\neg[p \vee [\neg p \wedge q]]$$

Counter-example: $\{p \mapsto \top, q \mapsto \perp\}$

$$p \wedge \neg p$$

Not only has a counter-example, but we can even prove negation.

$$\forall x . P(x)$$

Counter-example:

over the integers, when interpreting $P(x)$ as “ x is even”,
the formula certainly does not hold.

The negation $\neg \forall x . P(x)$ also does not hold.

Unsuccessful Proofs ...

What follows from this? nothing!

The formula either

- ▶ holds and we were not able to prove it, or it
- ▶ does not hold.

For showing that a formula does not hold we may

- ▶ find a **counter-example**, or
- ▶ even try to **prove** the **negation** of the formula.

Completeness



Kurt Gödel, 1906 (Brno) — 1978 (Princeton): Ph.D. thesis, 33 pages:

For every formula ϕ s.t. $\models \phi$, there is proof.

Then: ... in a different proof system [Gödel, 1929].

But it also holds for our system

(when including the equivalence between A and $\neg\neg A$)

Different Methods, Literature

Our method: informal version of *sequent calculus* [Ebbinghaus et al., 1984].

There are various *variants*

Alternative proof system: Hilbert calculus

Useful for studying the basics of mathematics, but
not for everyday proving.

Further Lectures

- ▶ Data structures: sets, lists, arrays etc.
- ▶ Program correctness
- ▶ Program correctness
- ▶ Program correctness
- ▶ Program correctness
- ▶ ...

Quantifiers, Equality, Equivalence: Example

$$\forall x \in \mathbb{N}. x \text{ is divisible by } 2 \Rightarrow 6x \text{ is divisible by } 4$$

We want to prove the implication for an arbitrary natural number, that is, a natural number, about which we do **not know anything**. So we introduce a **new** constant a , and prove

$$a \text{ is divisible by } 2 \Rightarrow 6a \text{ is divisible by } 4,$$

that is, we assume that a is divisible by 2 and prove: $6a$ is divisible by 4.

The definition of divisibility is:

$$\forall x, y. x \text{ is divisible by } y :\Leftrightarrow \exists k \in \mathbb{N}. yk = x$$

Especially:

- ▶ a is divisible by 2 $:\Leftrightarrow \exists k \in \mathbb{N}. 2k = a$
- ▶ $6a$ is divisible by 4 $:\Leftrightarrow \exists k \in \mathbb{N}. 4k = 6a$

So we can replace the formulas in the proof by equivalent ones

Quantifiers, Equality, Equivalence: Example Continued

From the assumption

$$\exists k \in \mathbb{N} . 2k = a$$

we prove

$$\exists k \in \mathbb{N} . 4k = 6a.$$

We give a name to the symbol k in the assumptions, e.g., k' , so we know

$$2k' = a.$$

We can write this equivalently as

$$12k' = 6a$$

$$4 \cdot 3k' = 6a$$

For proving the existential quantifier we choose for k the term $3k'$, so it suffices to prove

$$4 \cdot 3k' = 6a$$

which we already know. This finishes the proof.

Literature I

H.-D. Ebbinghaus, J. Flum, and W. Thomas. *Mathematical Logic*. Springer Verlag, 1984.

Kurt Gödel. *Über die Vollständigkeit des Logikkalküls*. PhD thesis, Universität Wien, 1929.